Wang (Vincent) Ng

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EDUCATION

University of California, Davis *Sept 2023 – Mar 2027*
BS in Computer Science, GPA: 3.73

- **Coursework:** Data Structures and Algorithm Design/Analysis, Computer Architecture, Intro to Artificial Intelligence, Operation Systems and Systems Programming, Object Oriented Programming, Databases, Linear Algebra
- **Involvement:** Google Developer Student Club, Codelab, Aggie Sports Analytics, AI Student Collective, Davis Undergraduate Engineering Network

WORK EXPERIENCE

Los Angeles Dodgers *Davis, CA*
Software Engineer – Part Time *Nov 2025 – Jun 2026*

- **Developed an automated pitching biomechanics analysis model** to extract arm angle and release point data from 200+ minor league videos with 90.67% accuracy to directly support the Dodgers' annual talent draft.
- **Applied skeletal tracking and physics simulation** to quantify pitcher performance metrics, enabling data-driven talent assessment and streamlining scout workflows for prospect evaluation.

Benevolent Bandwidth Foundation *Davis, CA*
Software Engineer – Contract *Jan 2026 – Jun 2026*

- **Built a full-stack delivery route optimization tool** in Next.js for small businesses (e.g. flower shops), enabling owners to plan up to 30-stop routes with a single click rather than manually sequencing deliveries.
- **Designed a session persistence layer** using sessionStorage to save and restore full optimizer state (vehicles, addresses, time windows) across page navigations with retry-on-refresh error recovery, ensuring no user data is collected.

PRODUCTS & PUBLICATIONS

Portfolio Diversification & Risk Assessment Analyzer *Jan 2026 – Jun 2026*

- **Developed a hierarchical K-Medoids clustering pipeline** to analyze all Russell 2000 stocks by market metrics and correlation to generate 10+ diversification recommendations.
- Integrated a Monte Carlo risk assessment module **simulating 100 price paths per stock** to compute VaR, CVaR, and drawdown.
- **Implemented a cloud scheduler** to automatically ingest daily Yahoo Finance market data, **keeping simulations and the interface updated with 5+ years of historical data.**
  - Tools: Python, BigQuery, NumPy, Pandas, scikit-learn, Next.js, TypeScript, React

LEANCODE: Context-Aware Codebase Simplification for LLMs *Jan 2026 – Mar 2026*

- **Engineered a batched attention extraction pipeline** over 908k Java training samples using fine-tuned CodeBERT and CodeT5 generative models, implementing CLS attention and encoder-decoder cross-attention from safetensors checkpoints with a custom 6-layer Transformer decoder added to CodeBERT for seq2seq generation.
- **Optimized AST-to-subword token alignment** from $O(W \times T)$ to $O(W + T)$ per sample via a single-pass character offset map, reducing per-sample alignment cost and enabling GPU-batched inference across the full training set.
- **Reproduced LEANCODE's context-aware token pruning** on the CodeSearchNet Java benchmark, achieving only 5.48% MRR drop and 10.17% BLEU-4 drop at 50% token removal.
  - Tools: Python, PyTorch, HuggingFace, CodeBERT, CodeT5, tree-sitter

Aggie Housing *Dec 2024 – June 2025*

- **Developed a housing discovery platform used by 50+ UC Davis students**, helping users identify apartments that best match their budget, location, and lifestyle preferences.
- **Managed and integrated 120+ apartment listings into Supabase**, linking key data (pricing, layouts, distance from campus) to a dynamic frontend interface.
- **Merged and analyzed 500+ reviews from Google and Yelp**, applying **NLP-based sentiment analysis** to automatically rank the best-rated apartments.
  - Tools: React, JavaScript, HTML/CSS, Flask, Python, Supabase, NLTK, Git

Windtrax *Mar 2025 – June 2025*

- **Designed and built a computer vision powered evaporative cooling fan** equipped with a front-mounted camera that activates upon detecting human motion.
- **Integrated and connected a pretrained YOLO v8-n model** to our fan to rotate the yaw towards a detected user up to 120º in each direction.
  - Tools: React, Python, JavaScript, GPIO, YOLO

SKILLS

- **Languages:** C++, C, Python, Java, JavaScript, TypeScript, HTML, CSS, Assembly
- **Frameworks:** React, FastAPI, Next.js, Node.js, Flask, Tailwind CSS, TensorFlow, PyTorch
- **Developer Tools:** Git, Figma, Google Colab, Supabase, Google Cloud, Jupyter Notebook
- **Libraries:** Pandas, NumPy, Matplotlib, Scikit-Learn, OpenCV, YOLO, NLTK